

ECSE 309 Formula Sheet

Waves

$$\omega = 2\pi f \text{ (rad/s)}$$

$$\beta = \frac{2\pi}{\lambda} \text{ (rad/m)}$$

$$\mu_p = f\lambda = \frac{dx}{dt} = \frac{\lambda}{T} \text{ (m/s)}$$

$$\lambda = \frac{c}{f} \text{ (m)}$$

$$F = \frac{1}{T} \text{ (Hz)}$$

$$A \cos\left(2\pi ft - \frac{2\pi}{\lambda} x\right) = A \cos(\omega t - \beta x)$$

Complex Numbers

$$e^{i\theta} = \cos \theta + j \sin \theta$$

$$ae^{i\theta} = x + jy$$

$$(x, y) = a(\cos \theta, \sin \theta)$$

$$a = \sqrt{x^2 + y^2}, \theta = \arctan\left(\frac{y}{x}\right)$$

Transmission Lines

Variable	Letter	Unit	Definition
Frequency	f	Hz	$\frac{u_p}{\lambda}$
Wavelength	λ	m	$\frac{u_p}{f} = \frac{\lambda_0}{\sqrt{\epsilon_r}}$
Wavelength in free space	λ_0	m	$\frac{c}{f}$
Angular Frequency	ω	rad/s	βu_p
Attenuation Constant	α	Np/m	
Phase Constant	β	rad/m	$\frac{\omega}{u_p}$
Permeability Support of mag. fields	μ	H/m	$\mu_r \mu_0$
Permittivity Support of elec. fields	ϵ	F/m	$\epsilon_r \epsilon_0$
Phase Velocity	u_p	m/s	$\frac{1}{\sqrt{\mu\epsilon}}$
Speed of light	c	m/s	$\frac{1}{\sqrt{\mu_0\epsilon_0}}$
Characteristic Impedance Impedance of infinite wire	Z_0, Z_c	Ω	$\sqrt{\frac{L'}{C'}}$
Load Impedance	Z_L	Ω	
Input Impedance	Z_{in}	Ω	$Z_0 \left(\frac{z_L + i \tan(\beta l)}{1 + iz_L \tan(\beta l)} \right)$
Generator Voltage	$\tilde{V}_g$	V	
Input Voltage	$\tilde{V}_{in}$	V	$\tilde{V}_g \frac{Z_{in}}{Z_g + Z_{in}} = V_0^+ + V_0^- = V(0)$
Voltage at a point	$V(x)$	V	$V_0^+ e^{-\gamma x} + V_0^- e^{\gamma x} = V_0^+ (e^{-\gamma x} + \Gamma e^{\gamma(x-2l)})$
Transmission line length	l	m	
Reflection coefficient at load	Γ		$\frac{Z_L - Z_0}{Z_L + Z_0} = \frac{z_L - 1}{z_L + 1} = \frac{V_0^-}{V_0^+} = -\frac{I_0^-}{I_0^+}$
Normalized Load Impedance	$z_L, \overline{Z_L}$		$\frac{1+\Gamma}{1-\Gamma} = \frac{Z_L}{Z_0}$
Standing Wave Ratio	S		$\frac{1+ \Gamma }{1- \Gamma }$
Forward and Backward Voltage wave phasor	V_0^+, V_0^-	V	$V_0^+ = \frac{V_{in}}{1+\Gamma e^{-2\gamma l}}$

Coaxial	2-Wire	Parallel-Plate
a inner radius	d wire diameter	w width of plate
b outer radius	D wire separation	h height of plate

Parameter	Coaxial	2-Wire	Parallel-Plate	Unit
R' Resistance	$\frac{R_s}{2\pi}(\frac{1}{a} + \frac{1}{b})$	$\frac{2R_s}{\pi d}$	$\frac{2R_s}{w}$	Ω/m
L' Inductance	$\frac{\mu}{2\pi}\ln\left(\frac{b}{a}\right)$	$\frac{\mu}{\pi}\text{arcosh}\left(\frac{D}{d}\right)$	$\frac{\mu h}{w}$	H/m
G' Conductance	$\frac{2\pi\sigma}{\ln\left(\frac{b}{a}\right)}$	$\frac{\pi\sigma}{\text{arcosh}\left(\frac{D}{d}\right)}$	$\frac{\sigma w}{h}$	S/m
C' Capacitance	$\frac{2\pi\epsilon}{\ln\left(\frac{b}{a}\right)}$	$\frac{\pi\epsilon}{\text{arcosh}\left(\frac{D}{d}\right)}$	$\frac{\epsilon w}{h}$	F/m

	Propagation Constant $\gamma = \alpha + i\beta$	Phase Velocity u_p	Characteristic Impedance Z_0
General case	$\gamma = \sqrt{(R' + i\omega L')(G' + i\omega C')}$	$\frac{\omega}{\beta}$	$\sqrt{\frac{R' + i\omega L'}{G' + i\omega C'}}$
Lossless $R' = G' = 0$	$\alpha = 0, \beta = \frac{\omega\sqrt{\epsilon_r}}{c}$	$\frac{c}{\sqrt{\epsilon_r}}$	$\sqrt{\frac{L'}{C'}}$
Lossless Coaxial	$\alpha = 0, \beta = \frac{\omega\sqrt{\epsilon_r}}{c}$	$\frac{c}{\sqrt{\epsilon_r}}$	$\left(\frac{60}{\sqrt{\epsilon_r}}\right)\ln\left(\frac{b}{a}\right)$
Lossless 2-wire	$\alpha = 0, \beta = \frac{\omega\sqrt{\epsilon_r}}{c}$	$\frac{c}{\sqrt{\epsilon_r}}$	$\left(\frac{120}{\sqrt{\epsilon_r}}\right)\text{arcosh}\left(\frac{D}{d}\right)$
Lossless parallel-plate	$\alpha = 0, \beta = \frac{\omega\sqrt{\epsilon_r}}{c}$	$\frac{c}{\sqrt{\epsilon_r}}$	$\frac{120\pi h}{\sqrt{\epsilon_r} w}$

$$\sqrt{\frac{\mu_0}{\epsilon_0}} \approx 120\pi\,\Omega$$

$$\text{arcosh}(a) = \ln(a + \sqrt{a^2 - 1})$$

$$a^2 \gg 1 : \text{arcosh}(a) \approx \ln(2a)$$

$$\text{Np} = 20 \log e \text{ dB}$$

$$\text{m/s} = 1.943844 \text{ knots}$$

$$V = IZ$$

Maxwell's Equations

Field	Meaning	
$\vec{D}$	Displacement	C/m ²
$\vec{E}$	Electric	V/m
$\vec{B}$	Magnetic	T
$\vec{H}$	Mag. Intensity	A/m
$\vec{J}$	Current Density	A/m ²

Law	Formula	Usage
Gauss	$\nabla \cdot \vec{D} = \rho_v$	$\oint_S \vec{D} \cdot d\vec{s} = Q$
Faraday	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint_C \vec{E} \cdot d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$
Gauss Mag	$\nabla \cdot \vec{B} = 0$	$\oint_S \vec{B} \cdot d\vec{s} = 0$
Ampere	$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	$\oint_C \vec{H} \cdot d\vec{l} = \int_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \cdot d\vec{s}$

$$\vec{D} = \epsilon \vec{E} \text{ in linear material}$$

$$C = \frac{Q}{V}$$

$$\vec{J} = \sigma \vec{E}$$